

Machine Learning for Natural Language Understanding

Prof. Achim Rettinger – Master NLP

Sources: Lecture

- Podcast: The Last Invention (EP1 – EP8)
 - <https://open.spotify.com/show/3JlvdUfyVkBoe9K1hiXWyl>
or
 - <https://podcasts.apple.com/us/podcast/the-last-invention/id1839942885>
or
 - https://www.youtube.com/playlist?list=PLH92J2PwyK2fmjlK6mXRLtWnDuB_glGCq

Flipped Classroom Questions

- What is Transhumanism and Posthumanism?
- Are you a Doomer, a Scout or an Accelerationist?
- What is the main driver behind AI development?
- Can AI systems be programmed to make ethical decisions?
- What are guardrails to reduce the risk of negative impacts of AI?



Q1: What is the main driver behind AI development?

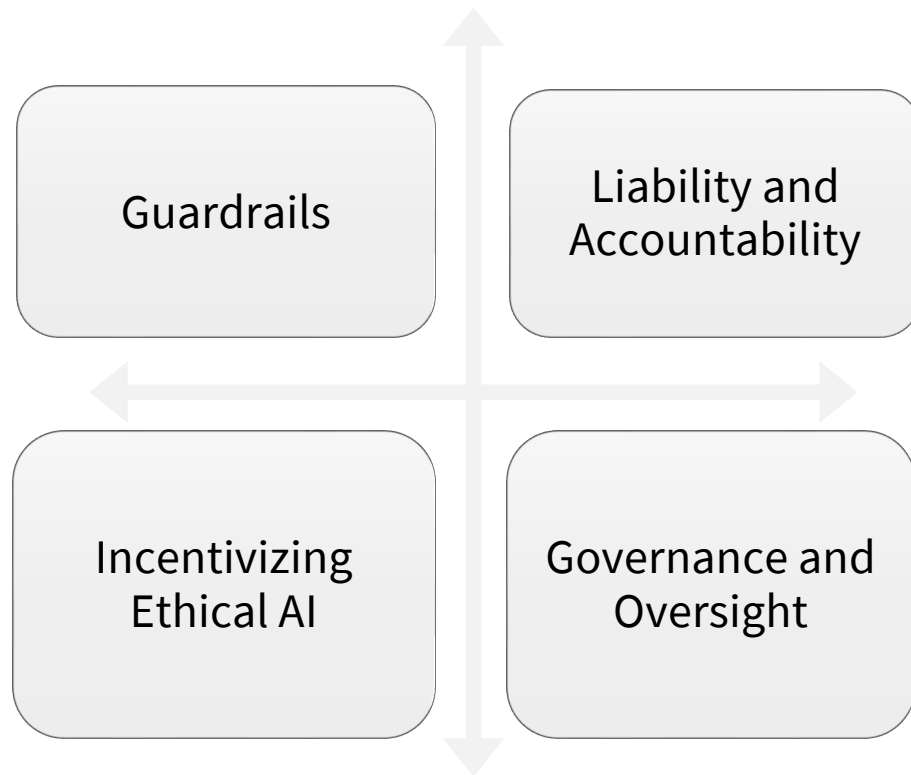
- Who is mostly pushing the research, development and use of AI now?
- What are the main reasons and motives?
 - Commercial interests
 - Technological evolution
 - Human curiosity
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Q2: Can AI systems be programmed to make ethical decisions?

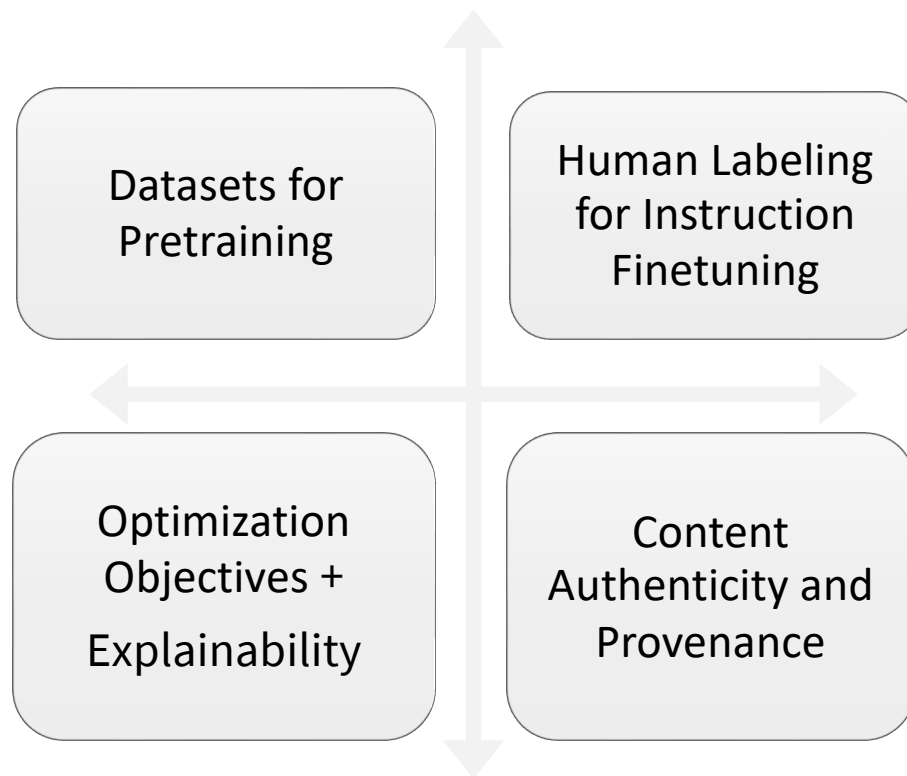
- Can you put law and jurisdiction into rules and code?
- Can morality be normalized and rationalized?
- Can any decision be objectively judged whether they are good or bad?

Countermeasures

Saving democratic processes



(Technical) Guardrails





The future of mankind

And what AI has to do with it

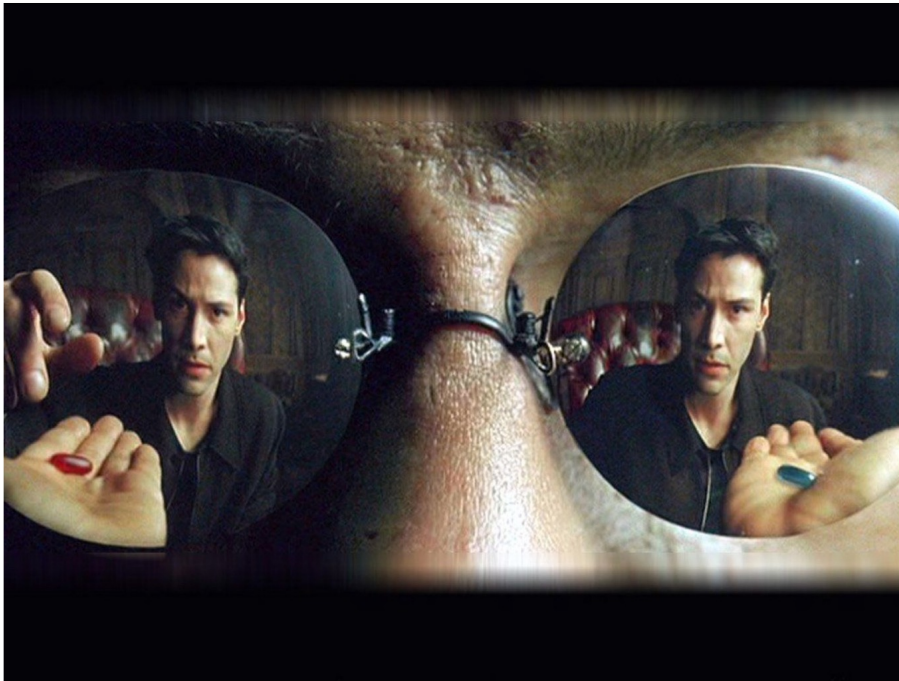
Dystopia: General Risks of AI

- Superhuman intelligence
- Intentional abuse of AI
- Dangerous use of AI without malicious intent
- Dangerous applications due to economic interests

Utopia: Paths for the survival of organized human life

1. Transhumanism -> Augment Humans
2. Posthumanism -> Replace Humans
3. Tech Optimism -> More efficient and effective technologies, standards of living keep improving, technology is applied well and solves climate crisis, hunger,...
4. Tech Pessimism / Ludits -> Reduced consumption of resources. Humankind does constrain itself and constrain the use of AI and technology.

Would you take the red or the blue pill?



Morpheus and Neo in The Matrix (1999)

If AI could fulfill all your wishes without you even realizing it, would you choose such a world of happiness, abundance and perfect security over our messy “real life”?



“The future of language is about the development of language models...”¹

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